

Rotating-Wheel LED Lights For Decoration At Functions

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This circuit produces a rotating-wheel like LED lights pattern, which can be used to decorate public halls during parties, festivals, and marriage functions. The concept is based on persistence of vision.

Circuit and working

The author's prototype is shown in Fig. 1 and the circuit diagram in Fig. 2. It is built around 12V-0-12V,

500mA step-down centre-tapped transformer (X1), two 1N4007 rectifier diodes (D1 and D2), 5V voltage regulator 7805 (IC1), NE555 timer (IC2), decade counter 4017B (IC3), and thirty 5mm LEDs (LED1 through LED30).

Regulated power supply for the circuit is derived from full-wave rectifier built using diodes (D1 and D2), transformer X1, 5V regulator 7805 (IC1), and two capacitors (C1 and C2).

The regulated 5V output of IC1 is used to drive the timer and counter circuits.

The 4017B (IC3) decade counter has ten outputs (Q0 through Q9). Each output goes high on receiving a clock input at its pin 14. Ten clock inputs are required to make

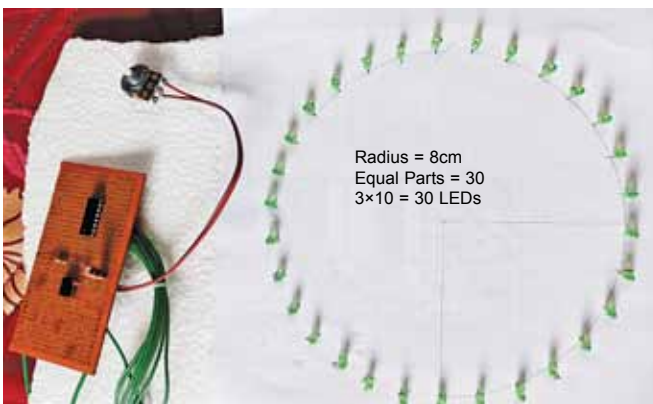


Fig. 1: Author's prototype

PARTS LIST

Semiconductors:

- IC1 - 7805, 5V voltage regulator
- IC2 - NE555 timer
- IC3 - 4017B decade counter
- D1, D2 - 1N4007 rectifier diode
- LED1-LED30 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 10-kilo-ohm
- VR1 - 10-kilo-ohm pot

Capacitors:

- C1 - 1000 μ F, 25V electrolytic
- C2 - 100 μ F, 16V electrolytic
- C3 - 10 μ F, 16V electrolytic
- C4 - 0.01 μ F ceramic disk

Miscellaneous:

- X1 - 230V AC, 50Hz primary to 12V, 500mA secondary transformer
- CON1 - 2-pin connector
- 11-pin female connector
- General-purpose PCB for LEDs
- Jumper wires
- Heat-sink for IC1
- 8-pin socket for IC2
- 16-pin socket for IC3
- Acrylic sheet (10cm $\times$ 10cm)

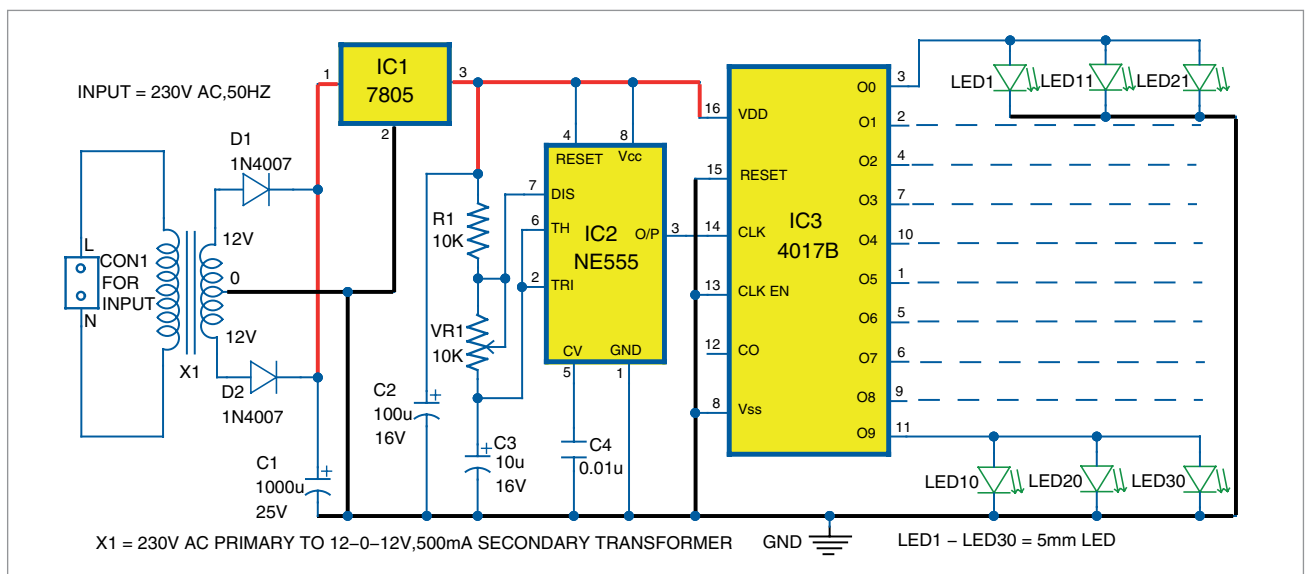


Fig. 2: Circuit diagram for the lights